

Where α hides

Charge unit, two spheres and the reference energy

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Where α hides

Abstract

FFGFT is first and foremost a ratio-based theory. It works in natural units with the convention $\alpha = 1$ and a correspondingly redefined charge; in this basic form it needs no fractal correction. This document shows where α goes under the redefinition and what geometric meaning it has there. With $\hbar = c = \epsilon_0 = 1$ one has $e^2 = 4\pi\alpha$; with $\alpha = 1$, $e = \sqrt{4\pi}$, and the charge is measured by the full surface of the sphere. α is then the square of the ratio of the historical to the geometric charge unit, and at the same time the ratio of two spheres around the electron, $\alpha = r_e/\lambda_C$. The FFGFT bridge $\alpha = \xi E_0^2$ reads $\xi E_0^2 = 1$ in T0 units, i.e. $E_0^2 = 1/\xi = 7500$. The only opaque element is the reference energy of the SI form $\alpha_{\text{SI}} = \xi(E_0/1 \text{ MeV})^2$: the agreement says that this reference energy is one MeV to within 10^{-4} . The MeV is not a geometric quantity but a unit of the SI chain, built from the historical charge unit in which α also hides. For the fractal correction it follows that in the α sector K_{frak} appears only in the translation into SI; the factor required by the two routes to α is the square of the bare reference energy in MeV.

Status markers

- [B]** Algebraically proven.
- [K]** Numerically confirmed (check script, comparison with measured values).
- [S]** Assumption, open or speculative.
- [X]** Refuted or not sustainable.
- [Q]** Convention.

Constants: CODATA 2022. Check script: `pruef_386_alpha_geometrie.py`.

1 The basic form of FFGFT

FFGFT calculates with ratios. In natural units, mass, energy, temperature and inverse time are one quantity in four guises (A085), and the coupling is adopted as a unit of calculation: $\alpha = 1$ **[Q]** (A267). The only parameter is $\xi = 4/30000$. In this basic form there is no fractal correction; K_{frak} , the SI values of α and G and the fully worked-out mass tables belong to the translation into SI, not to the core (A130). The question of this document is therefore not how precisely $1/137$ is reproduced, but where α goes when it is set to 1.

2 Where α goes under the redefinition

The fine-structure constant is

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}. \quad (1)$$

In natural units with $\hbar = c = \epsilon_0 = 1$ (Heaviside–Lorentz) what remains is

$$e^2 = 4\pi\alpha \quad \mathbf{[B]}. \quad (2)$$

The 4π is the full surface of the sphere. By Gauss's law the charge is the flux through a closed sphere; the flux is spread over the solid angle 4π . Setting $\alpha = 1$ gives

$$e_{\text{geo}} = \sqrt{4\pi} = 3.545, \quad (3)$$

so the charge is measured geometrically: its square is the full solid angle. With the measured α the historical charge is $e_{\text{hist}} = \sqrt{4\pi\alpha} = 0.3028$. Hence

$$\alpha = \left(\frac{e_{\text{hist}}}{e_{\text{geo}}} \right)^2, \quad \frac{e_{\text{geo}}}{e_{\text{hist}}} = \frac{1}{\sqrt{\alpha}} = 11.706 \quad \mathbf{[B]}. \quad (4)$$

Under the redefinition α therefore does not disappear. It moves into the charge unit: α is the square of the ratio of the historical to the geometric charge unit. In every calculation it stays on record through $e^2 = 4\pi\alpha$, one factor $\sqrt{\alpha}$ per vertex (A267).

The same 4π appears in the Higgs–vacuum route: the factor $64\pi^4 = 16\pi^3 \cdot 4\pi$ of the 2025 vacuum formula is the 4π from $4\pi\epsilon_0$ (Doc. 385).

3 The geometric picture: two spheres around the electron

Two spheres can be drawn around the electron. On the first, the Coulomb energy equals the rest energy; this is the classical electron radius

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2} = 2.818 \times 10^{-15} \text{ m.} \quad (5)$$

On the second, the quantum energy $\hbar c/r$ equals the rest energy; this is the reduced Compton wavelength

$$\lambda_C = \frac{\hbar}{m_e c} = 3.862 \times 10^{-13} \text{ m.} \quad (6)$$

Their ratio is exactly the fine-structure constant:

$$\frac{r_e}{\lambda_C} = \alpha, \quad \frac{a_0}{\lambda_C} = \frac{1}{\alpha} \quad \text{[B].} \quad (7)$$

Equivalently, α is the ratio of Coulomb energy to quantum energy on the same sphere, $(e^2/(4\pi\epsilon_0 r))/(\hbar c/r)$, for every radius r [K].

Geometrically, $\alpha = 1$ means that the Coulomb sphere and the Compton sphere coincide, and with them the Bohr radius; r_e , λ_C and a_0 are then a single scale $1/m_e$ (A267). The SI value $1/137$ only says that in the historical charge unit the Coulomb sphere is 137 times smaller than the Compton sphere. This is the transparent geometric meaning of α .

4 The FFGFT bridge in ratio form

FFGFT links α to ξ through the energy scale $E_0 = \sqrt{m_e m_\mu}$, the geometric mean of the two lightest charged leptons. In the basic form, i.e. in T0 units with $\alpha = 1$ and without fractal correction, the bridge reads

$$\xi E_0^2 = 1 \implies E_0^2 = \frac{1}{\xi} = 7500, \quad E_0 = 86.60 \quad \text{[B].} \quad (8)$$

The number $7500 = 1/\xi$ therefore belongs to E_0^2 , not to E_0 . Taken by itself, this equation fixes the T0 energy unit: one unit is $E_0/86.60 = \sqrt{\xi m_e m_\mu} = 84.85 \text{ keV}$ [K]. It acquires content only by comparison with a second scale, and that is the transition to SI.

5 The SI bridge and the reference energy

In SI the ratio form becomes (Doc. 011, A130)

$$\alpha_{\text{SI}} = \xi \left(\frac{E_0}{E_{\text{ref}}} \right)^2, \quad E_{\text{ref}} = 1 \text{ MeV.} \quad (9)$$

This is where the bridge becomes opaque. ξE_0^2 has the dimension energy², so a reference energy is needed, and the corpus takes the MeV. Computing backwards which reference energy the equation requires with the measured α gives

$$E_{\text{ref}} = E_0 \sqrt{\xi/\alpha} = \begin{cases} 0.99323 \text{ MeV} & \text{bare, } E_0 = 7.348 \text{ MeV,} \\ 0.99992 \text{ MeV} & \text{corrected, } E_0 = 7.397 \text{ MeV} \end{cases} \quad \text{[K].} \quad (10)$$

The bare value is $\sqrt{K_{\text{frak}}}$ MeV to within 8×10^{-5} . The fractal correction therefore enters only here, as a fine adjustment between the T0 unit and the MeV; without it $\alpha^{-1} = 7500/E_0^2 = 138.9$,

with it 137.06, 1.7×10^{-4} from the measured value. The same in the units of the basic form: the T0 energy unit is $\sqrt{\alpha} \cdot E_{\text{ref}}$.

The agreement of α thus states one thing: the reference energy is one MeV to within 10^{-4} . This is a property of the anchor E_0 (R135), not an independent prediction **[K]**. What the MeV is built from is shown in the next section.

6 Where the MeV comes from

The MeV is not a geometric quantity but a unit of the SI chain **[Q]**. One has

$$1 \text{ MeV} = 10^6 \cdot 1.602176634 \times 10^{-19} \text{ J.} \quad (11)$$

The numerical factor is the numerical value of the elementary charge in coulombs. Since 2019 it has been fixed; it was taken over from the old definition of the ampere via the force between two conductors, with $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$. The joule depends on the kilogram, metre and second. The MeV therefore carries the historical charge unit, exactly the unit in which α hides (Section 2).

The electron mass in MeV also has a measurement path in which α appears. What is measured spectroscopically is the Rydberg energy $Ry = 13.6057 \text{ eV}$, and

$$m_e c^2 = \frac{2 Ry}{\alpha^2} = 0.51100 \text{ MeV} \quad \mathbf{[B]}. \quad (12)$$

The SI form of the bridge, $\alpha = \xi m_e m_\mu / (K_{\text{frak}} \text{ MeV}^2)$, is equivalent to

$$\frac{m_e}{1 \text{ MeV}} = \sqrt{\frac{\alpha K_{\text{frak}}}{\xi (m_\mu / m_e)}} = 0.51104 \quad (+8 \times 10^{-5}) \quad \mathbf{[K]}. \quad (13)$$

In substance the relation thus fixes the ratio of the electron mass to the MeV through α , ξ and the mass ratio. The MeV itself is a chain of fixed numbers.

The two spheres of Section 3 show on which side the correction belongs. The Compton sphere $\lambda_C = \hbar / (m_e c)$ contains only the mass, i.e. the anchor. The Coulomb sphere $r_e = e^2 / (4\pi\epsilon_0 m_e c^2)$ contains e , i.e. the same historical unit from which the MeV is built. If the fractal correction is placed on this side instead of on the masses, the Compton sphere stays bare and the reference energy becomes

$$E_{\text{ref}} = \frac{E_0^{\text{bare}} \sqrt{\xi / \alpha}}{\sqrt{K_{\text{frak}}}} = 0.99992 \text{ MeV}, \quad (14)$$

i.e. 1 MeV to within 8×10^{-5} . Mathematically this is the same equation as with the corrected E_0 (R72); the reading is more coherent, however, because the masses stay bare and K_{frak} sits on the side of the unit, i.e. in the SI translation. This assignment is a choice **[S]**. The trail explains what the MeV is built from, not why it fits to within 10^{-4} ; the residual of 8×10^{-5} remains open **[S]**.

7 Where the fractal correction sits

The reference energy shows where K_{frak} sits in the α sector. The two routes to α in A130 differ only by this factor: route 1 inserts the bare mean $E_0 = \sqrt{m_e m_\mu}$ and gives $\alpha^{-1} = 138.9$, route 2 the corrected one and gives 137.06. The factor that α requires between the two is

$$K_\alpha = \xi m_e m_\mu \alpha^{-1} = \left(\frac{E_{\text{ref}}^{\text{bare}}}{1 \text{ MeV}} \right)^2 = 0.98650 \quad \mathbf{[B]}, \quad (15)$$

1.7×10^{-4} from 74/75 **[K]**. This is the same equation as in A130, only read differently, and therefore not a new measurement. It does show where the factor stands: K_α is the square of the bare reference energy measured in MeV, i.e. exactly the ratio between the T0 energy unit and the MeV. In the basic form it does not occur: there the bridge reads $\xi E_0^2 = 1$, which fixes the unit without leaving room for a correction factor **[B]**.

This agrees with Doc. 122: in ratios K_{frak} cancels, it acts only on absolute quantities measured against an external reference. For the α sector it follows that K_{frak} appears only in the translation into SI, as the adjustment between the T0 unit and the MeV **[K]**. Its value $74/75 = 1 - 100\xi$, by contrast, comes from the geometry, as the rotation number of the frozen recursion (Docs. 381, 384) and from the cumulative running (A040); origin and appearance are therefore separate. The residual of 1.7×10^{-4} between K_α and 74/75 remains open (R135, R139) **[S]**.

8 What holds and what is open

Statement	Status	Reason
$e^2 = 4\pi\alpha_i$ with $\alpha = 1$, $e = \sqrt{4\pi}$	[B]	$\hbar = c = \epsilon_0 = 1$
$\alpha = (e_{\text{hist}}/e_{\text{geo}})^2$	[B]	redefinition of the charge
$\alpha = r_e/\lambda_C$, $a_0/\lambda_C = 1/\alpha$	[B]	two spheres around the electron
$\alpha = 1$ as unit of calculation	[Q]	A267
$\xi E_0^2 = 1$, $E_0^2 = 1/\xi = 7500$ in T0 units	[B]	basic form
T0 energy unit $\sqrt{\xi} m_e m_\mu = 84.85 \text{ keV}$	[K]	check script
$E_{\text{ref}} = 0.99992 \text{ MeV}$ (corrected), 0.99323 MeV (bare)	[K]	property of the anchor (R135)
$1 \text{ MeV} = 10^6 \cdot 1.602 \times 10^{-19} \text{ J}$, numerical value of e	[Q]	SI chain, $\mu_0 = 4\pi \times 10^{-7}$
$m_e c^2 = 2 \text{ Ry}/\alpha^2$	[B]	measurement path of the electron mass
$m_e/\text{MeV} = \sqrt{\alpha K_{\text{frak}}/(\xi m_\mu/m_e)} = 0.51104$	[K]	$+8 \times 10^{-5}$, same residual
correction on the unit side, Compton sphere bare	[S]	reading; residual 8×10^{-5} open
$K_\alpha = (E_{\text{ref}}^{\text{bare}}/\text{MeV})^2 = 0.98650$	[B]	same equation as A130
K_{frak} in the α sector only in the SI translation	[K]	basic form $\xi E_0^2 = 1$; Doc. 122
residual 1.7×10^{-4} between K_α and 74/75	[S]	R135, R139

9 Aligned places in the corpus

With this document the places with $E_0 = 1/\xi$ carry a dated note pointing to the ratio form $E_0^2 = 1/\xi$: Doc. 005 (previously without a note), 032, 133 and 165, each De/En. Doc. 384 puts the basic form first. Doc. 013, like Doc. 014 before it, notes that $S_{T0} = 1 \text{ MeV}/c^2$ holds by construction. A130 notes that the gap between the two routes is the square of the bare reference energy.

10 Conclusion

α does not disappear under the redefinition; it becomes the charge unit. With $\alpha = 1$ the charge is measured by the full surface of the sphere, $e = \sqrt{4\pi}$, and the Coulomb sphere of the electron coincides with its Compton sphere. The measured value $1/137$ is the radius ratio r_e/λ_C in the historical charge unit. In its basic form the FFGFT bridge reads $\xi E_0^2 = 1$, i.e.

$E_0^2 = 1/\xi$; this is transparent. Only the step to SI is opaque, because it needs a reference energy that is one MeV to within 10^{-4} . The MeV is built from the numerical value of the historical charge, i.e. from the same unit in which α hides. On this side, and only there, the fractal correction sits: the Compton sphere stays bare, the value $74/75$ comes from the geometry, and it takes effect only in the adjustment to the MeV.

Check script: 2/python/Dok386_Skripte/pruef_386_alpha_geometrie.py — 21/21.